From Last Time...

A linear transformation is a function

$$T: \mathbb{R}^m \to \mathbb{R}^n, \quad T(\vec{x}) = A\vec{x}.$$

Rotations

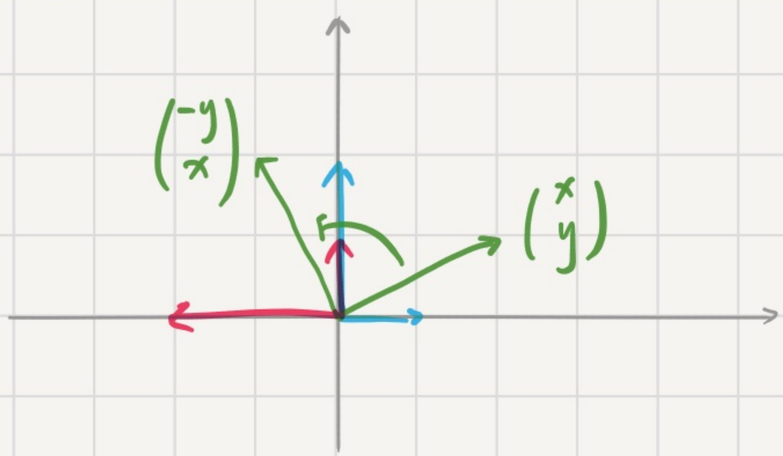
Ex. Recall:

$$T(\vec{x}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \vec{x}.$$

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

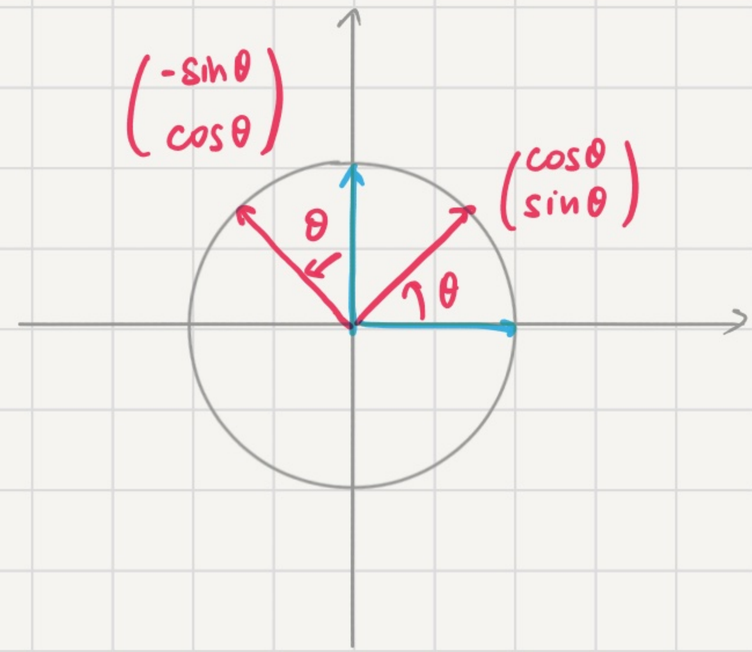
$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$



This is a 90° counter clockwise rotation.

Q. What is the standard matrix of a rotation by θ counterclockwise?



$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \leftarrow \text{first col. of } A$$

$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = A\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} \leftarrow \text{second col. of } A$$

The rotation matrix is

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

This is invertible: $R_\theta^{-1} = R_{-\theta}$.

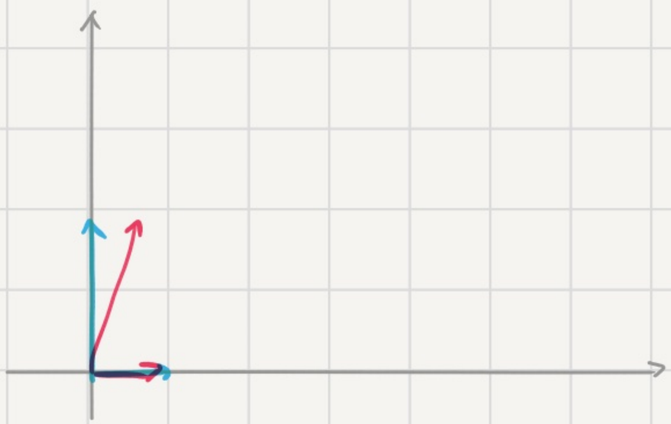
Shears

<u>Ex</u>. Consider

$$T(\vec{x}) = \begin{pmatrix} 1 & 0.2 \\ 0 & 1 \end{pmatrix} \vec{x}.$$

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0.2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$T\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 & 0.2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.4 \\ 2 \end{pmatrix}$$



The matrix of a horizontal shear is

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

For a vertical shear,

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}.$$

Geometric Transformations in $\mathbb{R}^2$

- Scaling $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$, $k > 0$
- Projections $\dfrac{1}{v_1^2 + v_2^2}\begin{pmatrix} v_1^2 & v_1 v_2 \\ v_1 v_2 & v_2^2 \end{pmatrix}$
- Reflections $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$, $a^2 + b^2 = 1$
- Rotations $R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$
- Shears $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$

A function is invertible if $T(\vec{x}) = \vec{y}$ has a unique solution x for all $\vec{y}$:

$$T^{-1}(\vec{y}) = \vec{x}.$$

A matrix $A_{n\times n}$ is invertible if the linear transformation $T(\vec{x}) = A\vec{x}$ is invertible.

* We require that A is $n \times n$.

<u>Ex</u>. Is the following matrix invertible?

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 8 & 2 \end{pmatrix}$$

Consider $T(\vec{x}) = A\vec{x}$:

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 8 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

goal: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \boxed{?} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$

$$\begin{aligned} x + y + z &= x' \\ 2x + 3y + 2z &= y' \\ 3x + 8y + 2z &= z' \end{aligned}$$

$$\left\{ \begin{aligned} x + y + z &= x' \\ 2x + 3y + 2z &= y' \\ 3x + 8y + 2z &= z' \end{aligned} \right.$$

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & x' \\ 2 & 3 & 2 & y' \\ 3 & 8 & 2 & z' \end{array}\right) \xrightarrow[R_3 - 3R_1]{R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & x' \\ 0 & 1 & 0 & -2x' + y' \\ 0 & 5 & -1 & -3x' + z' \end{array}\right)$$

$$\xrightarrow{R_3 - 5R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & x' \\ 0 & 1 & 0 & -2x' + y' \\ 0 & 0 & -1 & 7x' - 5y' + z' \end{array}\right)$$

$$\xrightarrow{-R_3} \left(\begin{array}{ccc|c} 1 & 1 & 1 & x' \\ 0 & 1 & 0 & -2x' + y' \\ 0 & 0 & 1 & -7x' + 5y' - z' \end{array}\right)$$

$$\xrightarrow{R_1 - R_2} \xrightarrow{R_1 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 10x' - 6y' + z' \\ 0 & 1 & 0 & -2x' + y' \\ 0 & 0 & 1 & -7x' + 5y' - z' \end{array}\right)$$

$$\Rightarrow \left\{ \begin{aligned} x &= 10x' - 6y' + z' \\ y &= -2x' + y' \\ z &= -7x' + 5y' - z' \end{aligned} \right.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underbrace{\begin{pmatrix} 10 & -6 & 1 \\ -2 & 1 & 0 \\ -7 & 5 & -1 \end{pmatrix}}_{\text{this is } A^{-1}!} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

$$
\begin{aligned}
x + y + z &= 1x' + 0y' + 0z' \\
2x + 3y + 2z &= 0x' + 1y' + 0z' \\
3x + 8y + 2z &= 0x' + 0y' + 1z'
\end{aligned}
$$

$$
\left(\begin{array}{ccc|ccc}
1 & 1 & 1 & 1 & 0 & 0 \\
2 & 3 & 2 & 0 & 1 & 0 \\
3 & 8 & 2 & 0 & 0 & 1
\end{array}\right)
$$

$$\downarrow$$

$$\vdots$$

$$\downarrow$$

$$
\left(\begin{array}{ccc|ccc}
1 & 0 & 0 & 10 & -6 & 1 \\
0 & 1 & 0 & -2 & 1 & 0 \\
0 & 0 & 1 & -7 & 5 & -1
\end{array}\right)
$$

$\hookrightarrow A^{-1}$

$$\boxed{\text{If } A \text{ is invertible, then } (A \mid I) \xrightarrow{\text{Elim.}} (I \mid A^{-1}).}$$

A Summary of Matrix Inverses

Let A be an $n \times n$ matrix.

Definition. A is invertible if

$T(\vec{x}) = A\vec{x}$ is invertible.

$$\Updownarrow$$

$A\vec{x} = \vec{y}$ has a unique solution for all $\vec{y} \in \mathbb{R}^n$.

Solving for $\vec{x}$:

$$(A \mid \vec{y}) \to \left(\underbrace{\boxed{\begin{matrix} 1 & & \\ & \ddots & \\ & & 1 \end{matrix}}}_{\text{Identity}} \;\middle|\; \begin{matrix} * \\ \vdots \\ * \end{matrix}\right)$$

$\Longleftrightarrow$

To find A^{-1}:

$$(A \mid I) \to (I \mid A^{-1}).$$

* If $\text{rref}(A) \neq I$, then A is not invertible.

<u>Theorem.</u> If A is invertible, then

$$AA^{-1} = I, \quad A^{-1}A = I.$$

If A, B are both invertible, then AB is also invertible:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Idea: $(B^{-1}A^{-1})AB = B^{-1}(A^{-1}A)B = B^{-1}B = I.$

$AB(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I.$